



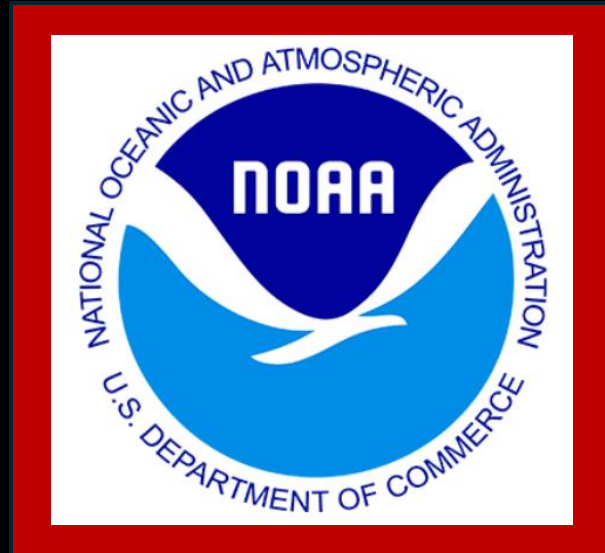
Project Plan

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Datasets

The dataset I am using for weather is one I requested from NOAA the National Oceanic and Atmospheric Administration. It consists of daily weather reports from all the weather stations in Montgomery County.

Important variables include latitude longitude and daily rainfall.

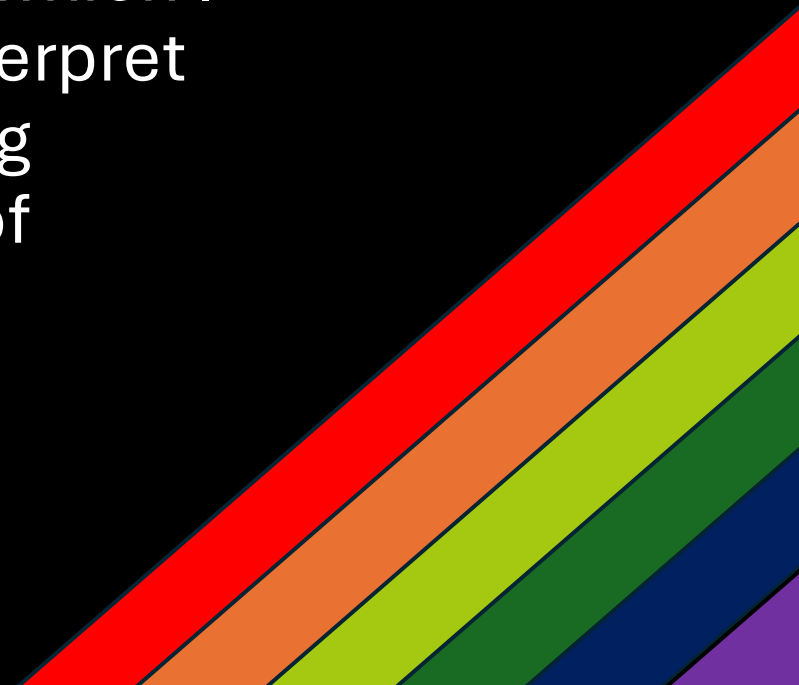


The dataset I am using for car crashes, is a record from data Montgomery. Important variables include latitude, longitude, and time.



Goals

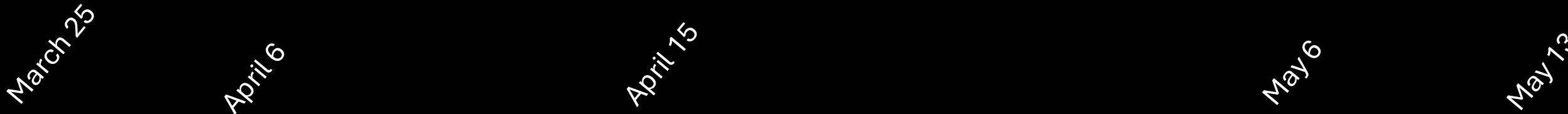
My goal continues to be to make a weather map overlaid with crash data. The main obstacle in the way of this is I don't have data for every single place in Montgomery County. To get around this I am going to attempt kriging which is the most eventful of my methods and thus the one I will talk about here. Kriging is a type of geospatial interpolation. Using kriging I can create what amounts to an image which I can overlay over a map created using leaflet. I will interpret this map to answer a multitude of questions regarding correlation between car crashes and rainfall or lack of correlation depending on what comes up. To perform kriging in R I will use a library called gstat.



Risks

- I have never done any geospatial interpolation and thus I still need to continue reading to figure out all the little things I can tweak to make the most accurate model I can.
- There is uncertainty in the method I am choosing to use.
- I can't really mitigate these problems but I can and will bring them up in my analysis as a sort of disclaimer

Project Time Line



Research and
practice
kriging

Create maps for specific days

Compile maps into timelapse



Depending on time do
more analysis

Project
progress
report

Finalize project